a Patient data stay in the hospital

Hospital network

Patient record

Open model, 4–9 B
one 40 GB GPU

evaluation and fine-tuning on site

Frontier APIs would move patient data outside the hospital

Model panel (native tool calling)

Qwen3.5-9B · Granite 4.1 8B · Gemma 4 E4B

b NeuroBench: one typed action per turn, scored against the case

Presentation

Man, 38 years
emergency department

Fever, headache and confusion for 18 h; ventricular shunt since infancy.

T 39.8 °C · HR 112

Nuchal rigidity; GCS 9.

Initial results

WBC 22.8 ×10⁹/L
CRP 218 mg/L
Procalcitonin 8.4 ng/mL

Local agent
one tool call or one assessment per turn

next turn

Action, turn 3
`order_ct_scan(contrast=false)`

Observation from the case $184
No mass lesion or midline shift; shunt catheter in place.

last turn

Final assessment
Bacterial meningitis; emergent; recommendations

16 diagnostic tools — results stored per case

MRI brain	CT head	Advanced imaging	Body imaging
EEG	ECG	Echo-cardiogram	Cardiac monitoring
Laboratory panels	CSF	Microbiology	Tissue diagnosis
Clinical assessment	Specialised tests	Literature	Drug interactions

imaging · physiology · laboratory · clinical

c Each case carries an executable policy reviewed by physicians

Case policy · BACT-MEN-RS02 — two independent physicians · seven dimensions

Required work-up

1. Blood cultures before antibiotics
2. CBC, CRP, procalcitonin, coagulation, HIV
3. CT head without contrast (GCS below 10)
4. CSF cells, protein, glucose, pathogens
5. MRI brain with contrast

Diagnosis

accepted bacterial, pneumococcal meningitis, …
vetoed viral, listerial, TB meningitis, abscess, …

Order

CT head → lumbar puncture **hard**
cultures → CT head soft

Avoid

ECG · echocardiogram · cardiac monitoring · FDG-PET

waste	lowers the waste score
harmful	caps the reward at 0.35
contraindicated	caps it at 0.10

Stop

No further tests after the required set.

d One verifiable reward for evaluation and for training

Policy reward R ∈ [0, 1]

diagnosis · required coverage · tool accuracy · safety · waste · sequence · stopping · cost · tokens

caps: contraindicated 0.10 · harmful 0.35 · hard order 0.50

evaluate · train

Benchmark score

Deterministic; no model judge.

40 test cases, conditions seen in training

60 test cases, two unseen conditions

Approved policies only.

RL on site (GRPO)

local model → rollouts on a training case → policy reward per episode → LoRA update → local model

e Accuracy and cost cannot see an unsafe or incomplete work-up

	Work-up of the same patient; all three diagnose bacterial meningitis — Actions in the order taken	Outcome metrics: Dx	Cost	Accuracy + cost rank	Policy metric: Policy reward	Policy rank
A	Blood cultures – Labs – CT head – Lumbar puncture – MRI complete, in order	✓	$1,313	2	0.95	1
B	Blood cultures – Labs – Lumbar puncture – CT head – MRI lumbar puncture before CT	✓	$1,313	2 tied	0.50	3
C	Labs – Lumbar puncture – Blood cultures – CT head – MRI cultures, CT and MRI skipped	✓	$682	1 cheapest correct	0.73	2